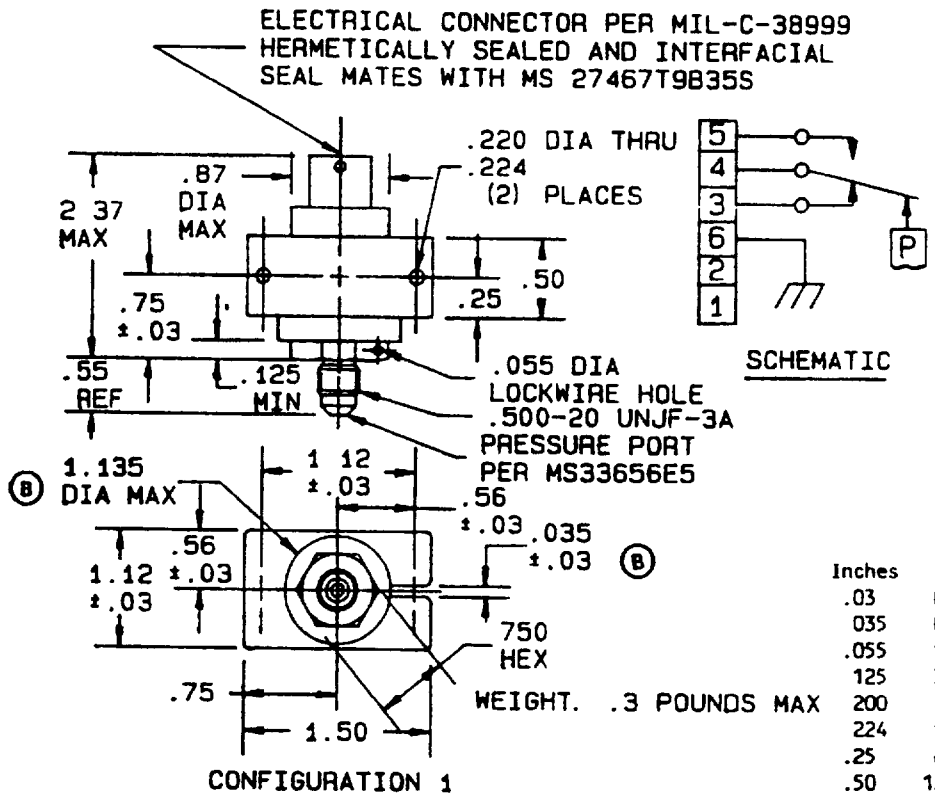
INCH-POUND
MIL-S-9395/39B
28 Feb 1992
SUPERSEDING
MIL-S-9395/39A
6 February 1979

MILITARY SPECIFICATION SHEET

SWITCHES, PRESSURE, (GAUGE) TYPE II, LOW LEVEL TO 5 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-9395



Inches	mm	Inches	mm
.03	0 8	.55	14.0
035	0 89	56	14.2
.055	1.40	750	19 05
125	3 18	.87	22 1
200	5 08	1.12	28.4
224	5 69	1 135	28.83
.25	6 4	1.50	38.1
.50	12.7	2 37	60.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Exact shape of switch is optional provided outline dimensions specified are not exceeded and mounting holes and connector location are as specified
4. Schematic shown is for switches with pressure ports exposed to zero lb_f/in^2
(B) 5. Unless otherwise specified, tolerances are ± 010 (0 25 mm) for three place decimals and ±.02 (0 51 mm) for two place decimals

FIGURE 1 Switches

(B) denotes changes

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:17Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions, configuration, and electrical schematic See figure 1

Weight See figure 1

Calibration See tables I, II, III, and IV.

System pressure 180 lb_f/in^2

Proof pressure 270 lb_f/in^2

Burst pressure 800 lb_f/in^2

Electrical ratings See table I.

Minimum current 25,000 cycles

Low level 50,000 cycles

NOTE Switches shall be subjected only to low level loads prior to delivery

Seal

Electrical chamber See table I

Pressure chamber Hermetic

Reference chamber Unsealed

Electrical connector See figure 1

Pressure port See figure 1

Media Dry air, nitrogen gas, fuel in accordance with MIL-T-5624, lubricating oil in accordance with MIL-L-7808, hydraulic fluid in accordance with MIL-H-5606 or MIL-H-83282, oxygen, or Coolanol 25R, or equal.

High temperature (operating and nonoperating) B (275°F)

Low temperature (operating and nonoperating) D (-65°F)

Altitude C (except 80,000 feet)

Shock: C (100 G)

Vibration S (test condition D, method 204 of MIL-STD-202), except 10 to 2,000 Hz, 20 G)

Supplemental nonoperating sinusoidal vibration

Sweep time 15 minutes

Frequency range and amplitude 50 through 81 Hz, 036 inch double amplitude, 81 through 210 Hz, ±12 G, 210 through 298 Hz, 0053 inch double amplitude; 298 through 500 Hz, ±24 G

Test duration 15 minutes in each of three mutually perpendicular planes

Life (mechanical) A (100,000 cycles)

Life (electrical) C (50,000 cycles)

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:17Z
Check the source to verify that this is the current version before use.

Acceleration C (8 G)

Pulsation amplitude A (0 percent)

Pulsation frequency. A (0 Hz)

Pressure rise A (less than 100 lb_f/in^2)

Dielectric withstanding voltage (at reduced barometric pressure). Applicable at 350 V rms

Electrical connector torque. 8 foot-pounds

Pressure port torque 15 foot-pounds

Terminal strength Applicable when terminals are used

Flame test. Applicable

Explosion Applicable

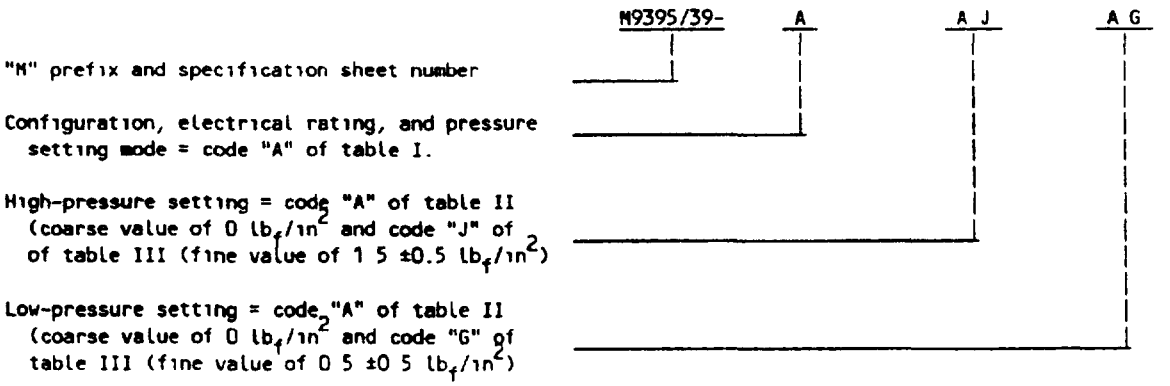
QUALIFICATION

Single submission Restricted to switch submitted

Group submission See table V

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash (-); and a five-letter code The five-letter code identifies the configuration, electrical rating, and pressure setting mode (code from table I), high-pressure setting (coarse value code from table II) followed by fine value with applicable tolerance (code from table III), and low-pressure setting (coarse value code from table II) followed by fine value with applicable tolerance (code from table III) The five-letter code used in the following example identifies a switch of configuration 1, low level to 1 ampere resistive at 28 V dc, which actuates on increasing pressure at 1.5 ±0.5 lb_f/in^2 and deactuates on decreasing pressure at 0.5 ±0 5 lb_f/in^2

Example



NOTE Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or acquired

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:17Z
Check the source to verify that this is the current version before use.

TABLE I Codes for combinations of configurations, electrical ratings, and pressure setting modes.

	Low level to 1 ampere resistive at 28 V dc		1 5 to 5 amperes at 28 V dc		Pressure setting mode	
	Configuration 1					
	Electrical chamber				High pressure	Low pressure
	Hermetic	Unsealed	Hermetic	Unsealed		
Code -	A	D	G	K	At (or max) 1/	At (or min) 1/
Code -	B	E	H	L	At (or max) 1/	Differential 2/
Code -	C	F	J	M	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III
2/ Setting values are designated by codes from table IV

TABLE II Codes for coarse settings.

Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)
A - - -	0	L - - -	30	W - - -	80
B - - -	2 5	M - - -	35	X - - -	85
C - - -	5	N - - -	40	Y - - -	90
D - - -	7 5	P - - -	45	Z - - -	95
E - - -	10	Q - - -	50	1 - - -	100
F - - -	12 5	R - - -	55	2 - - -	105
G - - -	15	S - - -	60	3 - - -	110
H - - -	17 5	T - - -	65	4 - - -	115
J - - -	20	U - - -	70	5 - - -	120
K - - -	25	V - - -	75	6 - - -	125

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:17Z
Check the source to verify that this is the current version before use.

TABLE III Codes for combinations of fine settings and tolerance values

	Fine value (lb_f/in^2) for settings below 20 lb_f/in^2					Tolerance (lb_f/in^2)
	0	.5	1	1.5	2	
Code - - -	A	B	C	D	E	±0 25
Code - - -	F	G	H	J	K	±0 5
Code - - -	L	M	N	P	Q	±1 0
Code - - -	R	S	T	U	V	±1 5
Code - - -	W	X	Y	Z	1	±2 0
Code - - -	2	3	4	5	6	±2 5
Code - - -	7	8	9	0	- 4/	Min or max

	Fine value (lb_f/in^2) for settings of 20 lb_f/in^2 and above					Tolerance (lb_f/in^2)
	0	1	2	3	4	
Code - - -	A	B	C	D	E	±1.0 1/
Code - - -	F	G	H	J	K	±2.0 2/
Code - - -	L	M	N	P	Q	±3.0 3/
Code - - -	R	S	T	U	V	±4 0
Code - - -	W	X	Y	Z	1	±5 0
Code - - -	2	3	4	5	6	±6 0
Code - - -	7	8	9	0	- 4/	Min or max

1/ Not applicable for pressure settings above 33 lb_f/in^2
2/ Not applicable for pressure settings above 66 lb_f/in^2
3/ Not applicable for pressure settings above 100 lb_f/in^2.
(B) 4/ A dash (-) is used as the code character for these fine setting and tolerance values.

TABLE IV Codes for differential settings 1/

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	T - - -	11
B - - -	5	U - - -	12
C - - -	1	V - - -	13
D - - -	1 5	W - - -	14
E - - -	2	X - - -	15
F - - -	2.5	Y - - -	16
G - - -	3	Z - - -	18
H - - -	3 5	1 - - -	20
J - - -	4	2 - - -	22
K - - -	4 5	3 - - -	24
L - - -	5	4 - - -	26
M - - -	5 5	5 - - -	28
N - - -	6	6 - - -	30
P - - -	7	7 - - -	35
Q - - -	8	8 - - -	40
R - - -	9	9 - - -	45
S - - -	10	0 - - -	50

1/ Differential settings require two codes, minimum differential and maximum differential

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:17Z
Check the source to verify that this is the current version before use.

TABLE V Extent of qualification

PIN	Number of samples required	Tests	Qualifies
Electrical chambers either hermetically sealed or unsealed			
M9395/39-GAJAG -AAJAG -A6R3W -G6R3W	2 each resistive 2 each intermediate current 2 each low level 2 each resistive	Complete in accordance with qualification inspection of MIL-S-9395	All
Electrical chambers unsealed			
M9395/39-KAJAG -DAJAG -D6R3W -K6R3W	2 each resistive 2 each intermediate current 2 each low level 2 each resistive	Complete in accordance with qualification inspection of MIL-S-9395	Configuration codes D, E, F, K, L, and M

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AV
Air Force - 99

User activities
Navy - AS, MC, SH
Air Force - 19

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5930-1484)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:17Z
Check the source to verify that this is the current version before use.